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## Article

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# Research on a Multi-Scale Tomato Ripeness Detection Method Based on an Enhanced YOLOv13s

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**Abstract:** Intelligent tomato harvesting robots are expected not only to satisfy basic harvesting needs but also to increase picking accuracy and efficiency, where tomato ripeness detection is especially important. This study investigates target recognition of tomato fruit ripeness under diverse cultivation environments. To improve detection accuracy and recall rate, an enhanced YOLOv13 algorithm (YOLOv13-SSW) is proposed. First, the SimAM attention mechanism is used to replace the last backbone layer of the YOLOv13s model. SimAM adaptively adjusts pixel weights by computing the similarity between each pixel and its neighboring pixels in the feature map, thereby highlighting salient features while suppressing irrelevant information. The DS-C3k2 convolution module in the backbone and neck structures is replaced with the DS-C3k2\_SAConv module. This module adopts SAConv to reduce spatial and channel redundancy during feature extraction and fusion, thereby increasing detection speed and decreasing model parameters. Finally, the loss function is optimized by replacing CIOU-Loss with WIOUv3 to accelerate convergence and improve convergence precision. Based on tests using 5,000 data samples over 300 epochs, the results show that the YOLOv13-SSW model achieves an accuracy of 91.612%, a recall rate of 89.678%, and an average precision of 92.319%. Compared with YOLOv13s, this corresponds to a 5.553% increase in accuracy, a 4.62% gain in recall rate, and an 8.033% rise in average precision. Comparative experiments with common models indicate that YOLOv13-SSW provides strong generalization ability, localization performance, detection performance, and robustness. Ablation experiments also confirm the necessity of each improvement, demonstrating that no redundant operations are introduced. These methods offer a useful reference for addressing similar tasks.

**Keywords:** Intelligent tomato harvesting robot; Object detection; Deep learning; Tomato fruit ripeness; Attention mechanism; Convolution method

## 1 Introduction

Tomatoes are among the most widely cultivated fruit and vegetable crops worldwide and consistently occupy the top position in annual global production. The tomato industry is also regarded as one of the three pillars of global vegetable trade, holding an important role in worldwide vegetable production. As the largest tomato-producing country, China accounts for about one-third of total global tomato output [1]. For many years, tomato harvesting in China has mainly depended on manual labor, leading to high labor costs and low operational efficiency. Workers usually assess tomato ripeness based on features such as color and texture. However, inconsistent ripening periods and subjective errors in manual picking greatly restrict the sustainable development of the tomato industry. At present, intelligent tomato harvesting machinery has become a major research direction, with strong emphasis on developing tomato ripeness recognition algorithms. Ongoing improvements in the accuracy and speed of tomato detection, recognition, and localization aim to realize efficient and intelligent harvesting operations [2].

With the rapid development of machine learning and neural networks, these technologies have been widely applied in target recognition because of their high detection precision, fast processing speed, and

strong generalization ability, especially for fruit recognition in natural environments [3]. Hernandez et al. proposed a tomato ripeness detection model based on YOLOv3-tiny, achieving an F1 score of 90% [4]. Touko et al. enhanced the YOLOv5 model to improve detection performance for obscured and overlapping tomatoes [5]. Miao et al. optimized YOLOv7 by substituting its backbone feature extraction network with lightweight MobileNetV3, reducing parameter computation while incorporating the GAM global attention mechanism to strengthen feature representation, which markedly improved accuracy [6]. Zheng et al. integrated the LSKA attention mechanism into the YOLOv8 model to enhance recognition of feature information across different positions. They proposed an ultra-lightweight and efficient dynamic sampling approach to replace traditional nearest-neighbor interpolation, enabling tomato fruit recognition in complex greenhouse scenarios [7]. Ma et al. combined the Ghost module with a coordinate attention module in YOLOv5 and introduced the CARAFE upsampling operator to build a feature fusion network, thereby enhancing tomato image detection performance [8].

Overall, object recognition models based on convolutional neural networks have been widely adopted in agricultural detection tasks. Nevertheless, current detection approaches still show several limitations. Tomato fruits often vary in size, and their quantities differ considerably, which frequently leads to stacking. Even within a single cluster, noticeable color differences remain. Leaf occlusion during harvesting and recognition is also unavoidable. These factors reduce the precision and stability of target recognition algorithms, resulting in inaccurate identification and slower processing speed. Tomato ripeness recognition algorithms deployed on intelligent tomato-picking robots also impose constraints on model size. At the same time, the specific characteristics of agricultural environments create practical challenges for YOLO-based algorithms. Previous studies indicate that even improved YOLO-series object recognition methods exhibit certain shortcomings in real applications. While recognition accuracy can be ensured, detection speed and model size often fail to satisfy practical demands. In contrast, emphasizing detection speed tends to reduce recognition accuracy.

Convolutional neural networks provide notable advantages for tomato ripeness detection research. In this work, YOLOv13, the latest model in the YOLO family, is selected as the baseline for tomato ripeness detection. It offers excellent real-time performance and can substantially improve harvesting robot operating efficiency. However, directly applying YOLOv13 cannot satisfy tomato fruit recognition requirements in complex scenes and shows limited operational efficiency when deployed on low-computing-power devices. Tomato plants typically bear many densely clustered fruits that strongly occlude each other. Adjacent fruits on the same plant also show different ripeness levels, and subtle fruit differences contribute to reduced recognition accuracy when using YOLOv13. To tackle these challenges, this study proposes a multi-scale tomato fruit ripeness recognition algorithm based on an improved YOLOv13s. By embedding the SimAM attention module into the YOLOv13 backbone network, feature extraction for green tomatoes under similar-color backgrounds is improved. In addition, SAConv convolutions are integrated into the original convolution structure between the backbone and neck networks to improve detection accuracy for small tomato targets and enhance feature fusion efficiency. The CloU loss function is replaced with WIoUv3-Loss to improve localization accuracy under occlusion and overlap. At the same time, the model training strategy is optimized. This method improves detection performance for tomato fruit maturity in complex conditions and significantly increases the picking efficiency and accuracy of intelligent tomato harvesting robots.

## **2 Introduction to YOLOv13**

### **2.1 YOLOv13 Network Architecture**

As the newest generation in the YOLO series, YOLOv13 is designed to improve real-time object detection performance and demonstrates strong capability, especially in complex scenes. Due to limitations in local feature aggregation and correlation modeling in YOLOv11 and YOLOv12, both detection accuracy and model size were adversely affected. YOLOv13 addresses these problems by introducing new mechanisms to capture more complex global and higher-order semantic correlations. The overall YOLOv13 network architecture still consists of three main components, including the Backbone network, the Neck network, and the Head network [9].

The Backbone section is responsible for feature extraction through a series of convolutional and deconvolutional layers. It incorporates residual connections and bottleneck structures to reduce network size while improving overall performance. YOLOv13 employs DSC3k2, A2C2f, and DSConv to extract features at different backbone stages. To decrease computational cost while preserving performance, YOLOv13 introduces Deep Separable Convolution (DSConv) to replace traditional large convolution kernels [10]. Deep Separable Convolution (DSConv [11]) decomposes the convolution operation into depthwise and pointwise convolutions, thereby markedly reducing model parameters and computational complexity. Compared with traditional convolutions, DSConv retains strong feature extraction capability while improving efficiency. To achieve lightweight optimization of the backbone architecture, DSConv is applied to improve the C3k2 module. Compared with the C3k2 module of YOLOv11, this optimization substantially decreases the number of parameters. The optimized backbone network allows YOLOv13 to achieve faster inference during runtime while maintaining high-precision object detection with low latency and reduced computational complexity.

The Neck component functions as a key hub for feature fusion and enhancement. YOLOv13 makes notable improvements and refinements in this section, mainly by integrating the HyperACE [12] mechanism and the FullPAD [13] paradigm. The HyperACE mechanism replaces traditional pairwise association modeling with hypergraphs. Unlike earlier approaches that rely on local information or simple self-attention, HyperACE learns multi-to-multi higher-order associations. It captures global semantic information across spatial locations and scales, making it especially effective for object interactions and close relationships in complex scenes. In contrast to fixed-threshold hypergraphs, the adaptive hypergraph computation in YOLOv13 dynamically learns the participation level of each pixel within hyperedges. This enables more flexible and accurate high-order relationship modeling, effectively capturing visual feature relationships across scales and locations. The FullPAD paradigm mainly strengthens information flow throughout the entire architecture, ensuring fine-grained coordination of features among the Backbone, Neck, and Head. By distributing enhanced features across the full pipeline, it improves gradient propagation, thereby increasing detection accuracy.

YOLOv13 shows leading performance on both the MS COCO and Pascal VOC datasets, achieving an average 1.5% mAP improvement over YOLOv12. Its Nano variant reaches 41.6% mAP with only 6.4 GFLOPS, and CPU inference speeds can reach up to 25 FPS. Under more complex scene conditions, small object detection exhibits noticeably higher stability.

In summary, YOLOv13 provides high detection accuracy and real-time performance while maintaining strong portability for deployment across different devices and environments. YOLOv13 is divided into four main versions: YOLOv13x, YOLOv13n, YOLOv13s, and YOLOv13l. The detection network depth and width of each version differ from previous iterations, enabling higher-dimensional and deeper-level detection to accommodate more complex scenarios. Given that detection accuracy, speed, and model size vary across versions, model selection should be determined by the research subject and detection objectives. This study aims to identify tomato fruit ripeness across diverse harvesting

conditions and requires deployment on an intelligent tomato-picking robot. Therefore, the model must provide strong real-time performance and reliable detection accuracy. Additionally, due to environmental variability, the inconsistency of tomato fruit size, and the influence of leaf occlusion, the model must satisfy strict performance demands. As a result, the model should not be excessively large while still ensuring sufficient accuracy. Therefore, YOLOv3s is selected as the baseline detection model for this research. Figure 1 presents the network architecture of YOLOv13s.

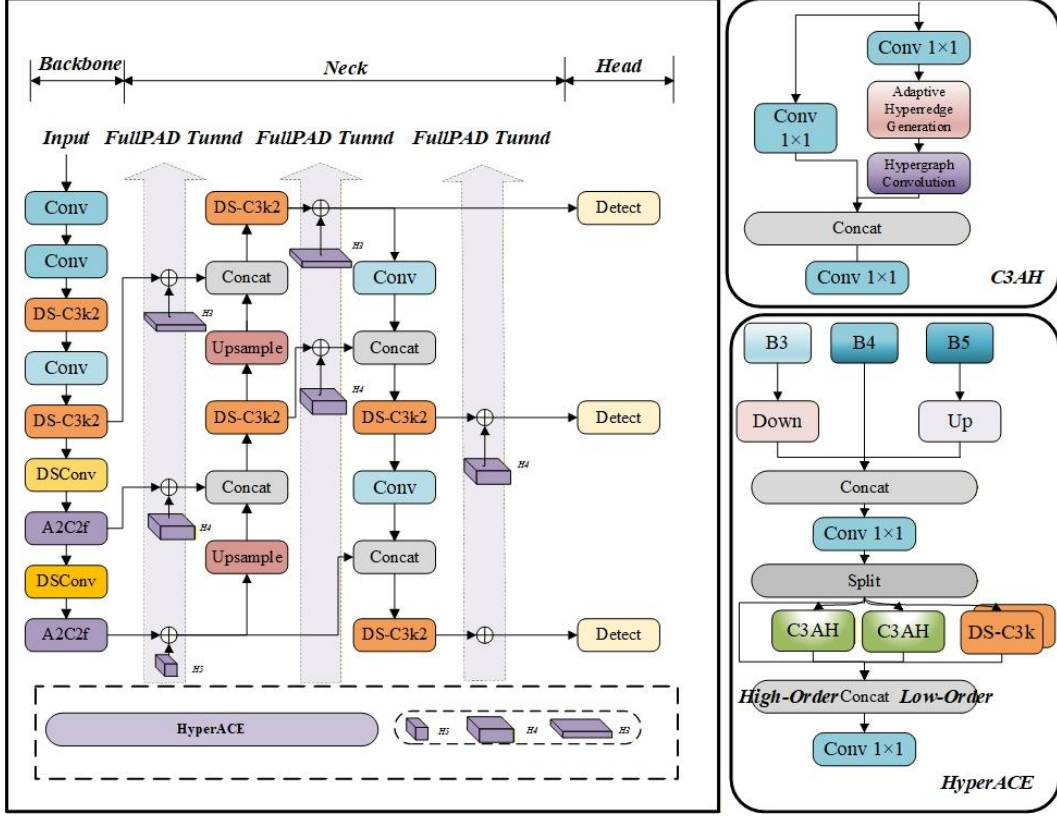


Fig. 1 YOLOv13s Network Architecture

### 3 Model Optimization and Improvement

#### 3.1 Attention Mechanism

Attention mechanisms, as an important component in machine learning, enhance model feature extraction capability while effectively suppressing background noise interference to improve robustness. They also improve detection accuracy for small and occluded objects, enabling more efficient use of features. This research focuses on recognizing tomato fruit ripeness and therefore requires stronger model attention. In complex harvesting environments, fruits may be blocked by leaves or by other fruits. Introducing an attention mechanism can effectively address this issue.

The attention module adopted in this study is SimAM (A Simple, Parameter-Free Attention Module for Convolutional Neural Networks) [14]. This attention mechanism differs from attention strategies used in existing models. SimAM can directly infer full 3D attention weights by jointly considering spatial and channel dimensions, rather than producing one-dimensional or two-dimensional weights along only the channel or spatial dimension. This allows the network to learn more discriminative features, better capture meaningful cues in images, and achieve improved consistency with image labels. The SimAM attention mechanism module is designed on the basis of neuroscience theory. It achieves attention by estimating the importance of individual neurons, inspired by visual processing mechanisms in the

mammalian brain, which provides stronger interpretability. Compared with most other modules that compute attention weights using heuristic strategies, the design of the SimAM attention module is more scientifically grounded and reasonable. Importantly, the SimAM attention module achieves its main feature—requiring no additional parameters in the original network—by deriving a closed-form solution for the energy function. This provides clear practical benefits, enabling lightweight implementation that substantially reduces model size without increasing complexity or computational burden, while effectively improving the performance of various convolutional neural networks across a wide range of visual tasks [15].

SimAM represents an attention mechanism on the basis of local self-similarity in feature maps. It dynamically adjusts the weight of each pixel by calculating the similarity between the pixel and its neighboring pixels, thereby strengthening key features and suppressing irrelevant ones. The main innovation of SimAM is its parameter-free design, allowing the model to achieve strong performance while maintaining low complexity. Its operation mainly includes the following steps, as schematically shown in Figure 2.

In visual neuroscience, the most informative neurons are typically those that exhibit firing patterns distinct from their neighbors. At the same time, an active neuron may inhibit activity in surrounding neurons, a phenomenon referred to as spatial inhibition.

Based on this principle, the SimAM attention mechanism defines an energy function for each neuron as expressed in Equation (1):

$$e_t(\omega_t, b_t, y, x_i) = (y_t - \hat{t})^2 + \frac{1}{M} \sum_{i=1}^{M-1} (y_0 - \hat{x}_i)^2 \quad (1)$$

Among them,  $\hat{t} = \omega_t t + b_t$  and  $\hat{x}_i = \omega_t x_i + b_t$  are linear transformation,  $t$  and  $x_i$  are the input features of  $X \in R^{C \times H \times W}$ , i.e., the target neuron and other neurons within a single channel.  $i$  refers to an index in the spatial dimensions,  $M = H \times W$  denotes the number of neurons in that channel.  $\omega_t$  and  $b_t$  indicate the weights and bias of the linear transformation.

To simplify the calculation, binary labels (i.e., 1 and -1) are used for  $y_t$  and  $y_0$ , and a regular term is added, resulting in the final energy function as expressed in Equation (2):

$$e_t(\omega_t, b_t, y, x_i) = \frac{1}{M} \sum_{i=1}^{M-1} (-1 - (\omega_t x_i + b_t))^2 + (1 - (\omega_t t + b_t))^2 + \lambda \omega_t^2 \quad (2)$$

By solving the above energy function, we obtain the solution for  $\omega_t$  and  $b_t$  are the closed-form solution is given by Equation (3):

$$\omega_t = -\frac{2(t - \mu_t)}{(t - \mu_t)^2 + 2\sigma_t^2 + 2\lambda}, b_t = -\frac{1}{2}(t + \mu_t)\omega_t \quad (3)$$

Here,  $\mu_t = \frac{1}{M-1} \sum_i x_i$  and  $\sigma_t = \sqrt{\frac{1}{M-1} \sum_i (x_i - \mu_t)^2}$  indicate the mean and variance of all neurons in the channel except (t).

Since the above solution is obtained on a single channel, assuming that all pixels within that channel follow the same distribution, the mean and variance can be computed once for all neurons and reused across that channel. The resulting minimum energy expression is presented in Equation (4):

$$e_t^* = \frac{4(\hat{\sigma}^2 + \lambda)}{(t - \hat{\mu})^2 + 2\hat{\sigma}^2 + 2\lambda} \quad (4)$$

Here,  $\hat{\mu} = \frac{1}{M} \sum_i x_i$  and  $\hat{\sigma}^2 = \frac{1}{M} \sum_i (x_i - \hat{\mu})^2$

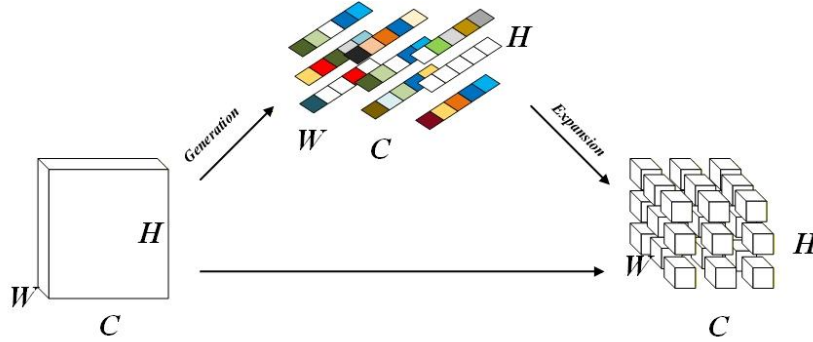
The lower the energy  $e_t^*$ , the greater the distinction between neuron t and its neighboring neurons,

and the more significant it becomes in visual processing. Thus, the importance of each neuron can be obtained through  $1/e_t^*$  this process.

In summary, the fundamental derivation of the energy function has been presented. Since attention modulation in the mammalian brain typically appears as a gain effect on neuronal responses, the SimAM attention mechanism module adopts scaling operators rather than addition for feature refinement. Equation (5) presents the refinement stage formula for the entire module:

$$\tilde{X} = \text{sigmoid}\left(\frac{1}{E}\right) \odot X \quad (5)$$

Here,  $E$  refers to the set of  $e_t^*$  values across all channels and spatial dimensions. The sigmoid function is used to prevent values in  $E$  from becoming excessively large. As a monotonic function, it does not change the relative importance of each neuron.



**Fig. 2** Schematic Diagram of the SimAM Attention Mechanism Principle

### 3.2 Convolution Method Improvement

This study analyzes the influence of factors such as tomato fruit size variation and ambient lighting conditions, which may cause omissions or misclassifications when the model extracts image information. Therefore, the convolution method was optimized accordingly.

This study adopts Switchable Atrous Convolution (SAConv) [16] to improve the convolution strategy of the original model. By incorporating SAConv, target information can be captured at a finer level through convolution operations with different atrous rates, thereby improving object detection performance. Integrating results from convolutions with different dilation rates provides a more complete feature representation. This allows the model to adapt to targets of different scales and offers an effective mechanism for converting pre-trained standard convolutional networks without retraining the entire model from scratch. Figure 3 presents the underlying principle.

SAConv is an improved and optimized extension of AConv (Atrous Convolution)[17]. AConv, also called atrous convolution, expands the filter field of view by inserting holes between elements in the convolution kernel, without increasing the number of parameters or computational complexity. For a void convolution with a void ratio of  $r$ ,  $r-1$  zeros are inserted between consecutive filter values, which is equivalent to expanding the filter kernel size of  $(k \times k)$  as shown in Equation (6):

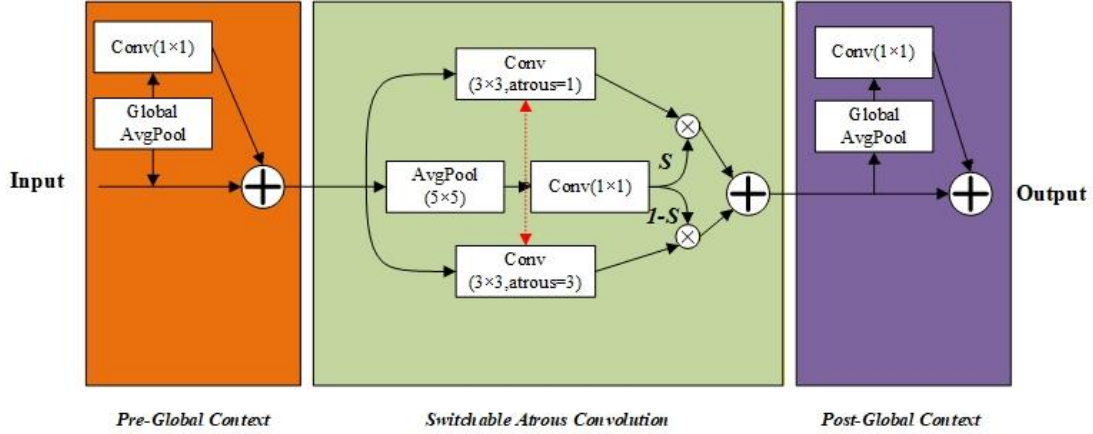
$$k_e = k + (k - 1)(r - 1) \quad (6)$$

The SAConv module applies convolution to the same input features using different dilation rates and uses switch functions to aggregate the results. These switch functions are spatially dependent, meaning each position in the feature map can have different switches controlling the SAConv output. This design allows the model to adaptively select the most suitable dilation rate for convolution results according to the target scale at different locations within the image. The SAConv module mainly includes

three parts: the central SAC component, with global context modules placed before and after it, as shown in the structural diagram of the SAConv module. The central SAC component acts as the core of the module and is responsible for converting the convolutional layer into SAC. For a convolutional layer, the transformation formula is shown in Equation (7):

$$\text{Conv}(x, \omega, 1) \xrightarrow{\text{toSAC convert}} S(x) \cdot \text{Conv}(x, \omega, 1) + (1 - S(x)) \cdot \text{Conv}(x, \omega + \Delta\omega, r) \quad (7)$$

where  $r$  represents a hyperparameter of the SAC, and  $\Delta\omega$  is a trainable weight. The switching function  $S(\cdot)$  is implemented by first applying an average pooling layer with an kernel, followed by a convolution layer. Two global context modules are inserted before and after the SAConv module. These modules first compress the input features using a global average pooling layer, and then feed the output back into the main stream through an convolution layer (without nonlinear layers). These modules resemble SENet but differ in aspects such as using only one convolution layer and applying different output processing methods. During implementation, all convolution layers in the backbone network (e.g., ResNet and its variants) are replaced with SAConv. Deformable convolution replaces the convolution operation in the formula, with its offset function initialized from predictions when loading from the pre-trained backbone network[18].



**Fig. 3** Schematic Diagram of SAConv Convolution Method Principle

### 3.3 Loss Function Optimization

Because the indoor target detection environment is more complex and variable, and contains more low-quality images or data, the regression loss function has a crucial impact on model performance. The YOLOv8 model uses CIOU-Loss as the regression loss function, but it does not perform well when handling low-quality data. Therefore, CIOU-Loss is replaced with the WIoUv3-Loss function, and the dynamic nonmonotonic focusing mechanism (WIoU) uses “outliers” instead of IoU to evaluate the quality of anchor frames, providing a reasonable strategy for assigning gradient gains. This method reduces the competitiveness of high-quality anchor frames while also lowering harmful gradients generated by low-quality examples. Since the quality of different anchor frames varies during detection, WIoU can better focus on normal-quality anchor frames, rather than forcing the model to concentrate only on high-quality anchor frames, thereby improving overall detection performance. Especially when the predicted box overlaps with the target box, WIoU can better guide the detection head to concentrate on target information. WIoUv3 constructs the distance-attention  $R_{WIoU}$ , which can effectively amplify the  $L_{IoU}$  of ordinary-quality anchor boxes, while  $L_{IoU}$  can significantly reduce the  $R_{WIoU}$  of high-quality anchor boxes. When the anchor boxes overlap well with the target box, it places greater emphasis on the distance between center points. This leads to the defining equation of WIoUv3:



$$L_{WIoUv3} = rR_{WIoU}L_{IoU} \quad (8)$$

where the defining equation of the  $R_{WIoU}$  distance attention and the defining equation of the LIoU loss function are, respectively:

$$R_{WIoU} = \exp\left(\frac{(x-x_{gt})^2 + (y-y_{gt})^2}{(W_g^2 + H_g^2)^*}\right) \quad (9)$$

$$L_{IoU} = 1 - IoU = 1 - \frac{W_i H_i}{S_u} \quad (10)$$

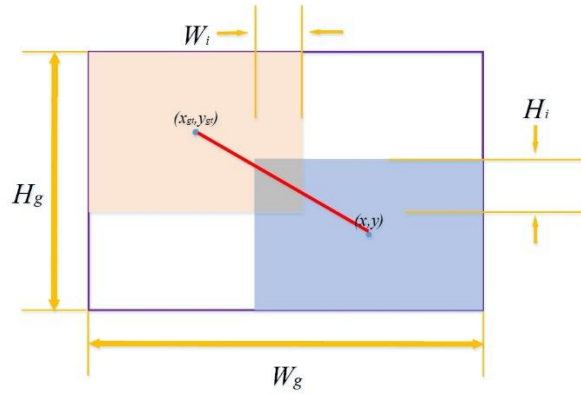
Low-quality anchor frame information is unavoidable during training, and factors such as the distance between the centroids of the prediction frame and the target frame, as well as the aspect ratio, all contribute to the prominence of low-quality anchor frame information, which reduces the overall detection accuracy of the model. The meaning of each parameter in Eq. (9) and Eq. (10) is shown in Figure 4. As shown in the figure, the blue box represents the ground truth box, the orange box represents the prediction box,  $W_i$  and  $H_i$  are the width and height of the intersection region of the two boxes,  $S_u$  is the area obtained from the union of the two boxes,  $x$  and  $y$  are the coordinates of the midpoints of the prediction box, and the width and height of the smallest box enclosing the two boxes are  $W_g$  and  $H_g$ , respectively.  $r$  in Eq. (8) is a non-monotonic focusing coefficient that can dynamically distinguish the quality of the anchor box. Its defining equation is shown in Eq:

$$r = \frac{\beta}{\delta \alpha^{\beta-\delta}} \quad (11)$$

Where  $\delta$  and  $\alpha$  are two hyperparameters, and  $\beta$  indicates the outlier degree. A smaller outlier degree indicates a higher-quality anchor frame, which is assigned a smaller gradient gain. In this way, attention to high-quality anchor frames is effectively suppressed, while BBR (Boundary Box Regression) is guided to focus on ordinary-quality anchor frames. This enables the model to emphasize global information, thereby improving overall generalization ability. At the same time, anomalously large values are also assigned a smaller gradient gain, which suppresses their contribution and reduces the impact on overall detection accuracy. Its defining formula is shown in Eq:

$$\beta = \frac{L_{IoU}}{\overline{L_{IoU}}} \in [0, +\infty) \quad (12)$$

Where  $\overline{L_{IoU}}$  refers to the exponential sliding average.



**Fig. 4** Schematic diagram of WIoUv3-Loss principle

### 3.4 Improved YOLOv13 Model

Given the complexity of the tomato harvesting environment in this project, along with variability in fruit ripeness, uncertainty in tomato size, and frequent leaf occlusion, the above optimization strategies were

integrated into the YOLOv13s model to construct an algorithm suitable for this study. The SimAM attention mechanism was introduced by replacing the final backbone section of the YOLOv13s model. SimAM dynamically adjusts the weight of each pixel by calculating the similarity between that pixel and its surrounding pixels in the feature map, thereby strengthening key characteristics and suppressing irrelevant ones. Next, the convolutional architecture in the YOLOv13s model was optimized. The DS-C3k2 convolution module in the backbone and neck structures was replaced with the DS-C3k2\_SAConv module, which applies SAConv to reduce spatial and channel redundancy throughout feature extraction and fusion. This improves detection speed and decreases the model parameter count. Finally, the loss function was refined by replacing CIOU-Loss with WIoUv3 to accelerate convergence speed and improve convergence accuracy. The integration of these enhancements ensures both detection accuracy and detection speed of the overall model. The improved model is named YOLOv13-SSW according to the adopted optimization approach. Figure 5 presents the network architecture of the YOLOv13-SSW model.

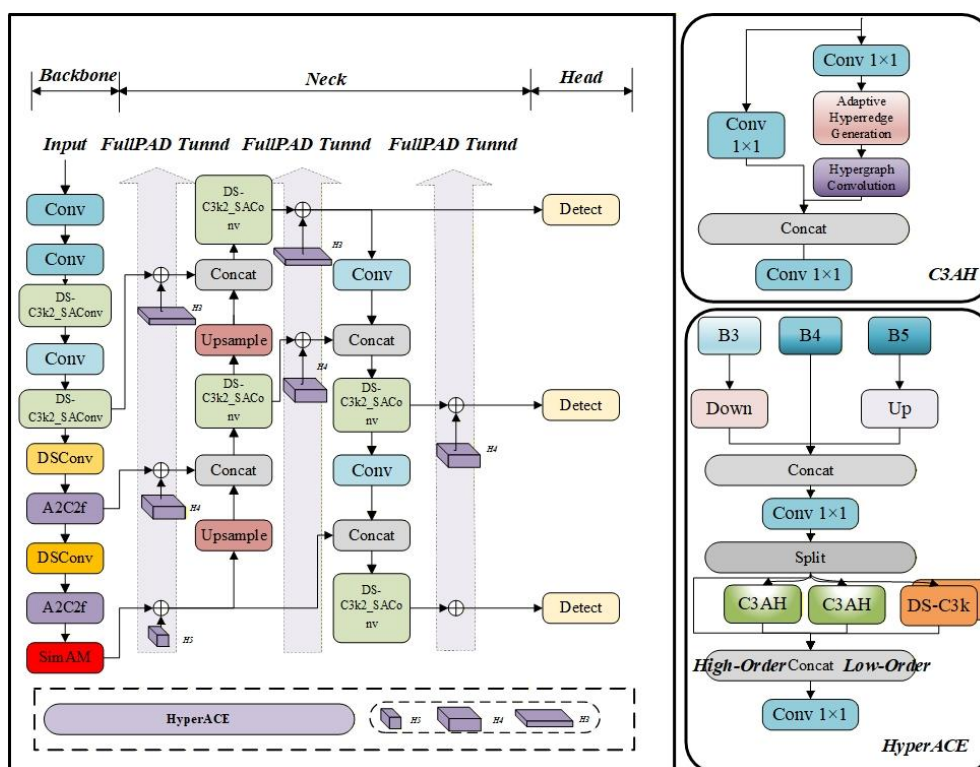


Fig. 5 YOLOv13-SSW Network Architecture Diagram

## 4. Establishment of the Dataset

Object detection algorithms are typically developed and encapsulated before being applied to specific domains. To detect particular targets, these algorithms must be trained and adapted into specialized models tailored to the target domain. However, the accuracy of the trained model directly determines the detection performance in subsequent applications. The training stage is the most critical step, making the quality and source of training data paramount. Therefore, constructing a comprehensive dataset that covers key influencing factors is essential for training object detection models with strong robustness and generalization ability.

### 4.1 Creation of the Dataset

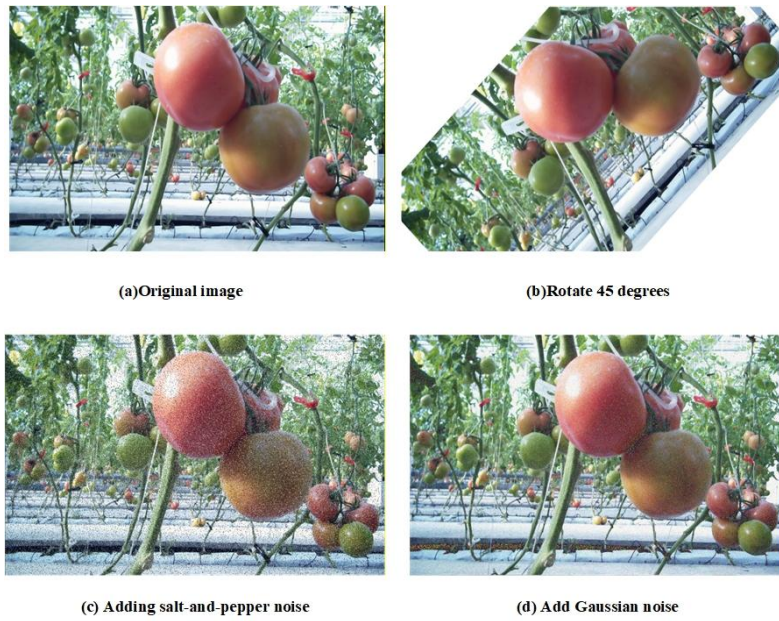
This project uses a tomato public dataset provided by the National Agricultural Data Center, together with the publicly available Tomato dataset and the FruitDet [19] dataset from the Roboflow repository. The National Agricultural Data Center dataset was collected in an intelligent tomato production greenhouse at the Jiangsu Academy of Agricultural Sciences. The greenhouse cultivated 18 rows of tomato varieties, including Sufen 11, with plants spaced 25 cm apart and a row spacing of 90 cm. Images were captured between December 2020 and February 2021 under mostly sunny conditions. During data acquisition, cameras were positioned 20–40 cm from the tomato plants, with tilt angles ranging from  $-45^\circ$  to  $45^\circ$ . Imaging was performed at 20 cm intervals along plant rows, capturing each row under diverse lighting scenarios, including front lighting, backlighting, and strong-light conditions. To enhance model generalization, multiple datasets were fused to construct a new self-built dataset for training. Figure 6 presents a subset of the dataset, including images with varying contrast, saturation, and degrees of fruit occlusion. This diversity effectively improves the robustness of the trained model.



**Fig. 6** Selected Images from the Dataset

## 4.2 Data Set Augmentation

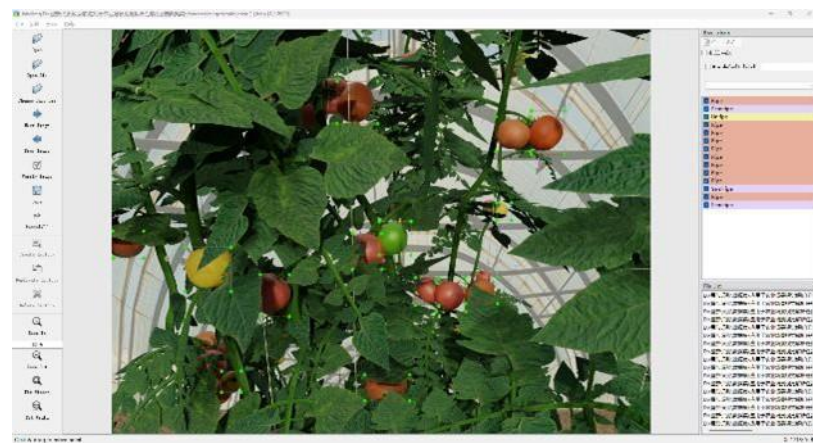
After constructing the dataset, to reduce issues such as overfitting during training, the image data were enhanced through operations such as noise addition and image flipping to expand the dataset, as shown in Figure 7, which illustrates the image preprocessing workflow [20]. The original dataset contained 1,000 images. After augmentation, the tomato fruit ripeness dataset increased to 5,000 images. Subsequently, the dataset was divided into training, validation, and test sets at an 8:1:1 ratio to evaluate overall model performance.



**Fig. 7** Image Preprocessing

### 4.3 Annotation of the Dataset

Based on the newly constructed tomato fruit ripeness dataset, which consisted entirely of unlabeled images, the dataset required for model training needed to be annotated. Therefore, the self-built tomato fruit ripeness dataset was labeled using the LabelImg [21] software, as shown in Figure 8. This study primarily focuses on detecting and identifying tomato fruit ripeness under complex and variable environmental conditions. Accordingly, the annotation categories were divided into three classes: Ripe, Semiripe, and Unripe. The annotation files were saved in .xml format. Figure 9 presents an example annotation file, clearly indicating the location and class name of the annotated object within the image, together with other relevant information.



**Fig. 8** Example of Image Annotation



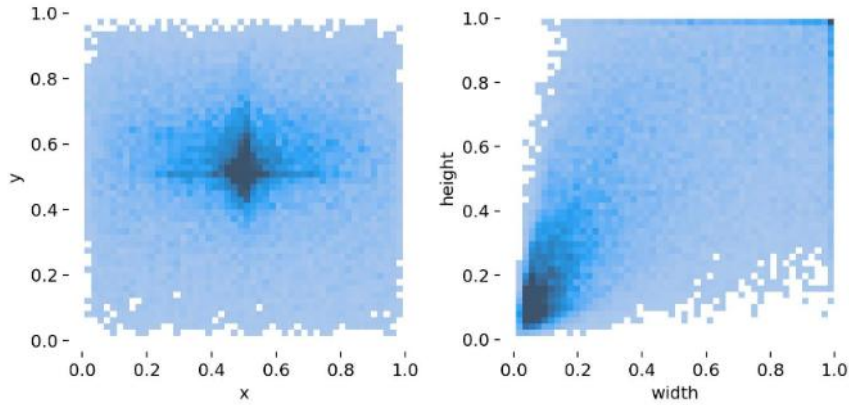
```

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2   <folder>train</folder>
3   <filename>color (1).bmp</filename>
4   <path>D:\番茄识别\数据集\应用于农业场景视觉解析任务的番茄数据集\datasets\images\train\color (1).bmp</path>
5   <source>
6     <database>Unknown</database>
7   </source>
8   <size>
9     <width>1600</width>
10    <height>1200</height>
11    <depth>3</depth>
12  </size>
13  <segmented>0</segmented>
14  <object>
15    <name>Ripe</name>
16    <pose>Unspecified</pose>
17    <truncated>0</truncated>
18    <difficult>0</difficult>
19    <bndbox>
20      <xmin>1140</xmin>
21      <ymin>259</ymin>
22      <xmax>1241</xmax>
23      <ymax>362</ymax>
24    </bndbox>
25  </object>
26  <object>
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35      <ymax>777</ymax>
36    </bndbox>
37  </object>
38  <object>
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41    <truncated>0</truncated>
42    <difficult>0</difficult>
43    <bndbox>

```

**Fig. 9** Example of a Labeling File Diagram

To improve the model generalization ability, it is essential to ensure the completeness of the dataset. Specifically, the positions and categories of detected objects should be distributed across all regions of the images. Therefore, after visualizing the annotated dataset to examine the spatial distribution of labeled objects, the label distribution map shown in Figure 10 was obtained. The horizontal axis  $x$  represents the ratio of label width to image width, whereas the vertical axis  $y$  denotes the ratio of label height to image height [22]. The results indicate that the label data in the dataset are relatively comprehensive, with broad coverage across different positions and object sizes. This distribution is consistent with practical application scenarios for tomato maturity detection in complex environments, where small- to medium-sized targets are predominant.



**Fig. 10** Label Distribution Map

## 5 Experimental Results and Analysis

To validate the feasibility of the proposed model, relevant experiments were conducted. The experimental

platform configuration included Windows 11, an Intel® Core™ i5-12400F CPU @ 2.4 GHz (8 CPUs), ~2.4 GHz; an NVIDIA GeForce RTX 4060 Ti GPU; and 32 GB of RAM. PyTorch 1.10.1 was used as the deep learning framework. To accelerate deep learning computations, CUDA 10.8 and cuDNN 8.0.5 were employed. Model training was prioritized, and the training parameters are listed in Table 1.

**Table 1** Training Parameters

| Parameter name   | Parameter value |
|------------------|-----------------|
| Momentum         | 0.937           |
| Weight decay     | 0.0005          |
| Batch size       | 8               |
| lr               | 0.01            |
| epoch            | 600             |
| Warm up epoch    | 3.0             |
| Warm up Momentum | 0.8             |
| Warm up bias lr  | 0.1             |

## 5.1 Evaluation Indicators

To verify the performance improvement of the YOLOv13 object detection algorithm model, the following metrics are used as evaluation indicators: precision (P), mean average precision (mAP), recall (R), model size, number of model parameters, and frames per second (FPS). The relevant evaluation metrics are expressed as follows:

$$P = \frac{TP}{TP+FP} \times 100\% \quad (13)$$

$$R = \frac{TP}{TP+FN} \times 100\% \quad (14)$$

$$\begin{cases} AP = \int_0^1 P dR \\ mAP = \sum_1^n \frac{AP_i}{n} \end{cases} \quad (15)$$

Among these, FN indicates the number of positive samples incorrectly predicted as negative during model detection, FP represents the number of negative samples incorrectly predicted as positive during detection, TP represents the number of positive samples correctly predicted as positive during detection, and AP represents the area under the P-R curve, which is the average precision. n represents the number of categories in the dataset used for model detection.

## 5.2 Comparative Experiments

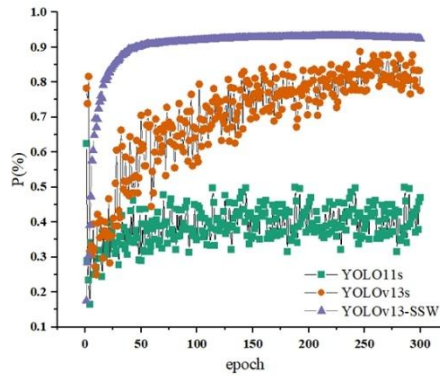
To verify the object detection performance of the improved model, comparative experiments were conducted between the improved model and currently mainstream object detection algorithms. The selected comparison models include all versions of YOLOv13 (n, s, l, x), YOLOv5, YOLOv8, YOLO11, and Faster R-CNN. The comparison experiments used the same tomato fruit ripeness dataset for training and validation. The experimental results are compared in Table 2, which presents detection metrics across different models. Precision (P), Recall (R), mAP@0.5 (mAP), mAP@0.5–0.95 (mAP), and computational resource requirements (GFLOPs) were used as evaluation metrics for model detection performance.

**Table 2** Comparative Experimental Results

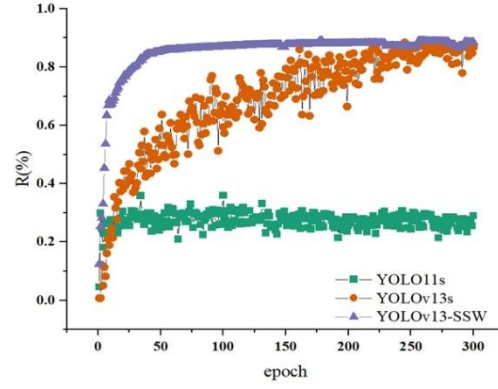
| Model       | P              | R              | mAP@0.5        | mAP@0.5 – 0.95 | GFLOPs    |
|-------------|----------------|----------------|----------------|----------------|-----------|
| Faster-RCNN | 83.271%        | 72.518%        | 80.127%        | 79.623%        | 67        |
| YOLOv5      | 84.624%        | 67.281%        | 76.583%        | 73.924%        | 51        |
| YOLOv8      | 85.037%        | 74.453%        | 81.083%        | 80.013%        | 38        |
| YOLO11      | 84.938%        | 78.945%        | 83.692%        | 81.862%        | 46        |
| YOLOv13n    | 86.824%        | 85.210%        | 85.721%        | 86.991%        | 37        |
| YOLOv13x    | 85.518%        | 83.254%        | 83.428%        | 84.528%        | 49        |
| YOLOv13l    | 87.628%        | 85.420%        | 86.589%        | 87.358%        | 51        |
| YOLOv13s    | 86.059%        | 85.058%        | 84.286%        | 85.397%        | 36        |
| YOLOv13-SSW | <b>91.612%</b> | <b>89.678%</b> | <b>92.319%</b> | <b>90.035%</b> | <b>20</b> |

As shown in Table 2, the improved YOLOv13s model, YOLOv13-SSW, achieves higher precision (P), recall (R), and mean average precision (mAP) than the other models. At the same time, it requires fewer parameters than other models, resulting in faster detection speed. It can also be observed that other YOLOv13 variants show improved performance compared with YOLOv11. Therefore, YOLOv13 is better suited to deliver stronger performance in tomato fruit ripeness detection. Based on the comparative experiments, YOLOv13-SSW is confirmed as the target detection algorithm model for this research project.

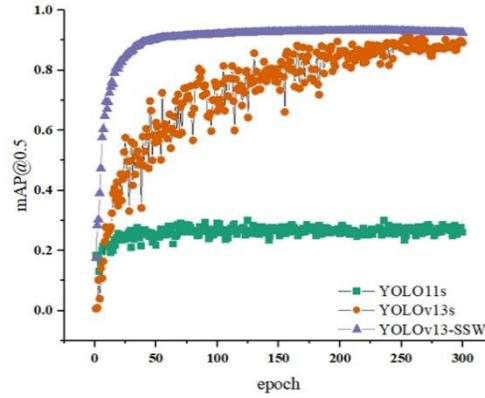
To visually compare differences in model performance, the widely used and stable YOLOv11 model was selected as one benchmark. Since YOLOv13-SSW is developed from YOLOv13s, YOLOv13s was also selected as a comparison model. Figure 11 presents the comparison curves of P, R, and mAP values among typical models. It can be seen that YOLOv13-SSW achieves the best and most stable performance with faster convergence. At the same time, the training results of all three models show some instability. This is mainly due to uncertainties such as variations in tomato fruit color and size during ripening, as well as leaf occlusion. As a result, fluctuations remain in the training process. However, the final trend and regression outcomes confirm the stability and usability of the model. In addition, the YOLOv13s model shows better performance than YOLO11. Therefore, for optimizing existing application domains, the YOLOv13 series models can be promoted as suitable emerging models for implementation. Overall, the YOLOv13-SSW model achieves high accuracy, high recall, and relatively strong stability.



(a)



(b)

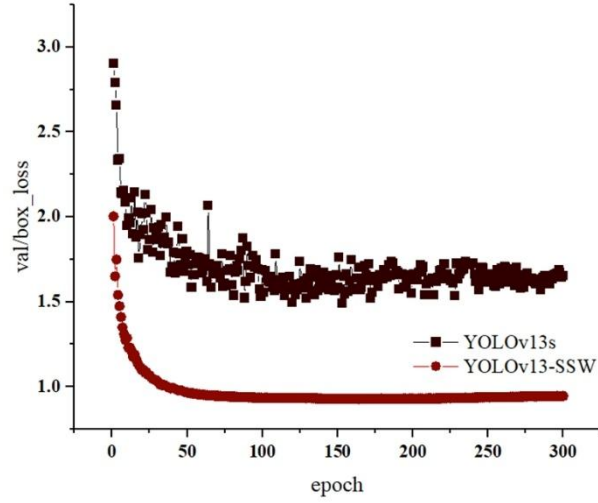


(c)

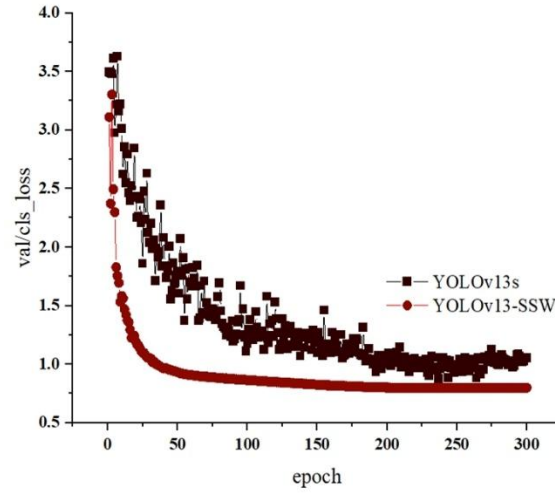
**Fig. 11** Comparison of P, R, and mAP Values for Typical Models

To compare model convergence speed and regression performance under different loss functions, the improved model YOLOv13-SSW was compared with the original model YOLOv13s. Figure 12 presents the loss function comparison curves between the models, where Figure (a) shows the comparison of bounding box loss function curves. During model detection and the pre-labeled calibration boxes, the bounding box loss function measures the error between the generated prediction boxes. The results indicate that YOLOv13-SSW converges faster and reaches better convergence, while YOLOv13s shows instability in bounding box generation and weaker convergence. Figure (b) shows the comparison of category loss function curves, which evaluates whether the predicted detection categories match the pre-labeled classification categories. Similarly, YOLOv13-SSW shows better convergence speed and performance, whereas YOLOv13s still shows poor convergence. Figure (c) presents the comparison of object loss function curves. The object loss function mainly evaluates whether the predicted grid contains the target categories to be detected. Likewise, YOLOv13-SSW shows stronger overall convergence performance. In summary, the results indicate that YOLOv13-SSW achieves faster convergence with improved convergence quality. Its loss functions show stable trends and stronger regression capability. This confirms the effectiveness and reliability of the model improvements.

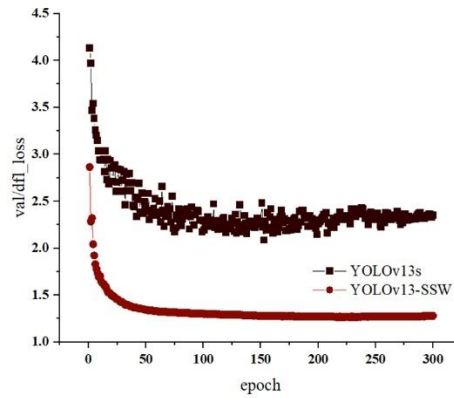




**Fig. 12-(a)** Comparison Curve of Target Box Loss Functions between Models



**Fig. 12- (b)** Comparison Curve of Category Loss Functions between Models



**Fig. 12--(c)** Comparison Curve of Target Loss Functions between Models

### 5.3 Ablation Studies

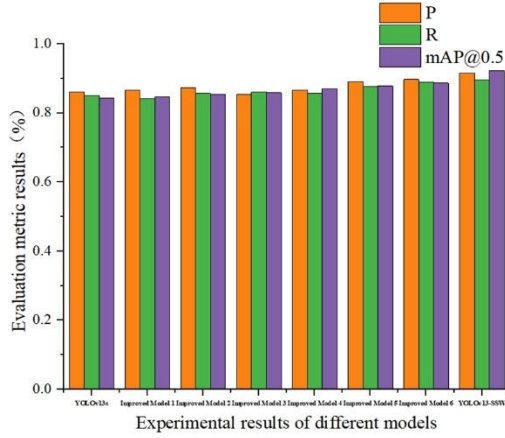
To verify the effectiveness of the three enhancements added to YOLOv13 and to determine the contribution of specific components, ablation studies were conducted by removing certain elements from the overall network architecture and observing changes in model performance, thereby confirming the rationality of structural optimizations. The ablation experiments were performed on the baseline model

YOLOv13s. First, to incorporate the SimAM attention mechanism, the DSC3k2 module in the Backbone and Neck sections was replaced. Next, the convolution strategy in the YOLOv13s model was optimized by replacing the DSC3k2 convolution module in the Backbone and Neck structures with the DSC3k2\_SAConv module, using SAConv convolutions. Finally, the model loss function was modified by replacing WIoUv3-Loss with CIoU-Loss. These improvements are summarized in Table 3, which reports ablation experiment results evaluated using P, R, and mAP@0.5–0.95 as performance metrics.

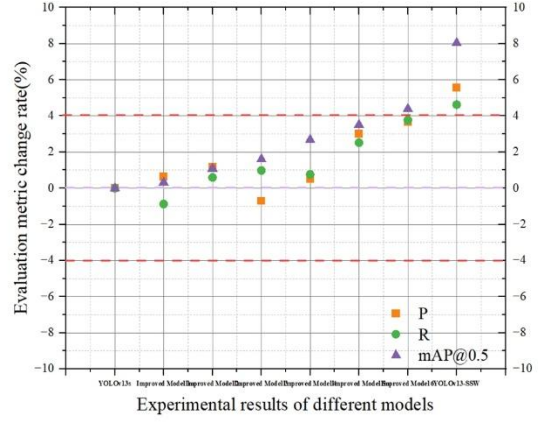
**Table 3** Ablation Experiment Results

| Model            | SimAM | SAConv | WIoUv3-Loss | P              | R              | mAP@0.5        |
|------------------|-------|--------|-------------|----------------|----------------|----------------|
| YOLOv13s         | —     | —      | —           | 86.059%        | 85.058%        | 84.286%        |
| Improved Model 1 | ✓     | —      | —           | 86.683%        | 84.186%        | 84.602%        |
| Improved Model 2 | —     | ✓      | —           | 87.249%        | 85.647%        | 85.376%        |
| Improved Model 3 | —     | —      | ✓           | 85.354%        | 86.038%        | 85.884%        |
| Improved Model 4 | ✓     | ✓      | —           | 86.583%        | 85.816%        | 86.958%        |
| Improved Model 5 | ✓     | —      | ✓           | 89.059%        | 87.573%        | 87.793%        |
| Improved Model 6 | —     | ✓      | ✓           | 89.728%        | 88.831%        | 88.671%        |
| YOLOv13-SSW      | ✓     | ✓      | ✓           | <b>91.612%</b> | <b>89.678%</b> | <b>92.319%</b> |

Table 3 shows that analysis of the ablation experiments reveals the following results. Improved Model 1 only adds the SimAM attention mechanism to the original YOLOv13s model, achieving a 0.624% increase in precision and a 0.316% improvement in mean average precision, while causing a 0.872% decrease in recall. Improved Model 2 improves the convolution method of the original model by replacing it with SAConv. This yields a 1.19% increase in detection accuracy, a 0.589% rise in recall rate, and a 1.09% improvement in mean average precision. Improved Model 3 optimizes the loss function by replacing CIoU-Loss with WIoUv3-Loss. This results in a 0.705% decrease in detection precision, a 1.598% increase in mean average precision (MAP), and a 0.98% increase in recall. Improved Model 4 combines the SimAM attention mechanism and SAConv convolution in the original model, achieving a 0.524% increase in precision, a 0.758% improvement in recall rate, and a 2.672% increase in mean average precision (MAP). Improved Model 5 integrates the SimAM attention mechanism with the WIoUv3-Loss function into the original model, achieving a 3% increase in precision, a 2.515% increase in recall, and a 3.507% increase in mean average precision (MAP). Improved Model 6 integrates the SAConv convolution method and WIoUv3-Loss into the original model, achieving a 3.669% increase in precision, a 3.773% increase in recall, and a 4.385% increase in mean average precision (MAP). Finally, integrating all three improvements into the original model produces the enhanced model YOLOv13-SSW, achieving a 5.553% increase in precision, a 4.62% rise in recall, and an 8.033% improvement in mean average precision (MAP). In summary, YOLOv13-SSW shows the strongest overall improvement. While other enhancements provide gains, they also lead to partial losses. This confirms the effectiveness and reliability of the proposed model improvements. The superiority of the improved model YOLOv13-SSW across all metrics is clearly shown by the ablation experiment analysis presented in Figure 13. Therefore, the enhanced model can be used as the object detection model for this research project.



(a) Bar Chart of Ablation Experiment Evaluation Metrics



(b) Scatter plot of changes in evaluation indicators for ablation experiments

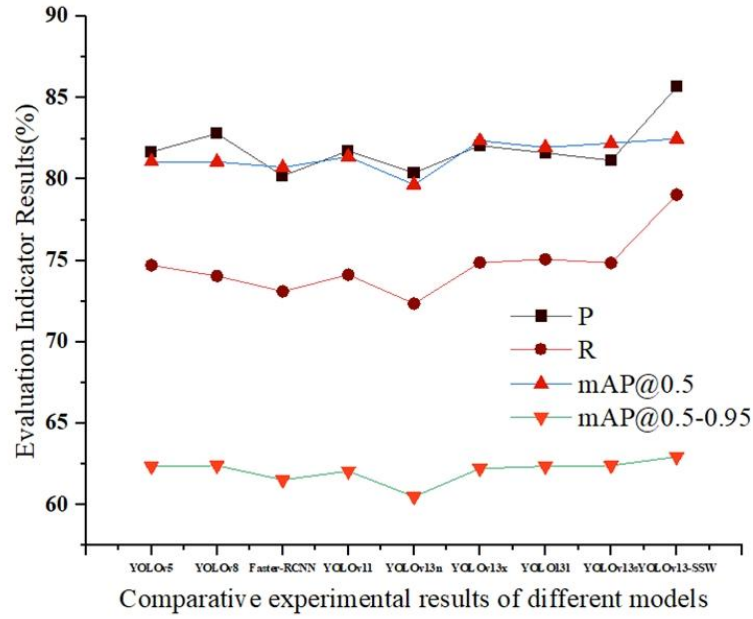
Fig. 13 Analysis Chart of Ablation Experiment Results

#### 5.4 Model Generalization Experiment

This study conducted experiments on the public VOC2007 dataset and compared it with different models to comprehensively verify the generalization capability of the improved model. The model training parameter settings are shown in Table 1 to remain consistent with the above experiments, and the evaluation metrics remain unchanged. The experimental results from training and validation are presented in Section 4. To present the results more clearly and intuitively, comparison diagrams were created. The relevant results are provided in Figures 14.

Table 4 VOC2007 Comparative Test Results

| Model       | P%           | R%           | mAP@0.5%     | mAP@0.5-0.95% | Detection time/ms |
|-------------|--------------|--------------|--------------|---------------|-------------------|
| YOLOv5      | 81.69        | 74.72        | 81.09        | 62.37         | 57                |
| YOLOv8      | 82.81        | 74.06        | 81.07        | 62.42         | 67                |
| Faster-RCNN | 80.21        | 73.12        | 80.73        | 61.54         | 108               |
| YOLOv11     | 81.75        | 74.14        | 81.38        | 62.08         | 37                |
| YOLOv13n    | 80.39        | 72.36        | 79.67        | 60.51         | <b>32</b>         |
| YOLOv13x    | 82.07        | 74.88        | 82.37        | 62.24         | 52                |
| YOLO13l     | 81.62        | 75.08        | 81.96        | 62.37         | 76                |
| YOLOv13s    | 81.16        | 74.86        | 82.22        | 62.42         | 51                |
| YOLOv13-SSW | <b>85.67</b> | <b>79.06</b> | <b>82.49</b> | <b>62.95</b>  | 35                |



**Fig. 14** Schematic diagram of experimental results comparing different models on the VOC2007 dataset

Comparative experiments on public datasets show that YOLOv13-SSW achieves the best evaluation metrics, although the differences among models are small. While there is still room for improvement, the advantages of the model across all metrics are clearly demonstrated. This confirms that the improved model has strong generalization capability and robustness, making it suitable not only for detecting tomato ripeness targets in complex environments but also for broader application across different scenarios.

### 5.5 Visualization Results and Analysis

Test samples were selected from the dataset that presented challenging detection conditions to demonstrate the robustness and generalization capability of the improved YOLOv13 model. These included samples with harsh picking environments, low light levels, severe tomato fruit occlusion, dense tomato distribution, distant positions, and far-away targets, all of which are prone to misclassification. Samples with similar ripeness levels were also included for verification. Figure 15-(a) shows annotated images from the test set. Figure 15-(b) presents the detection results from YOLOv13s, which clearly shows false negatives and false positives (default). Missed detections and misclassifications often occur when tomatoes are heavily occluded, small in size, or have similar maturity levels. Figure 15-(c) shows that the detection results of YOLOv13-SSW are substantially improved compared with YOLOv13s. The issues of missed detections and false positives are effectively reduced. The model achieves strong detection performance for tomato fruit ripeness in various complex harvesting environments, while also markedly improving the confidence scores of detected objects.



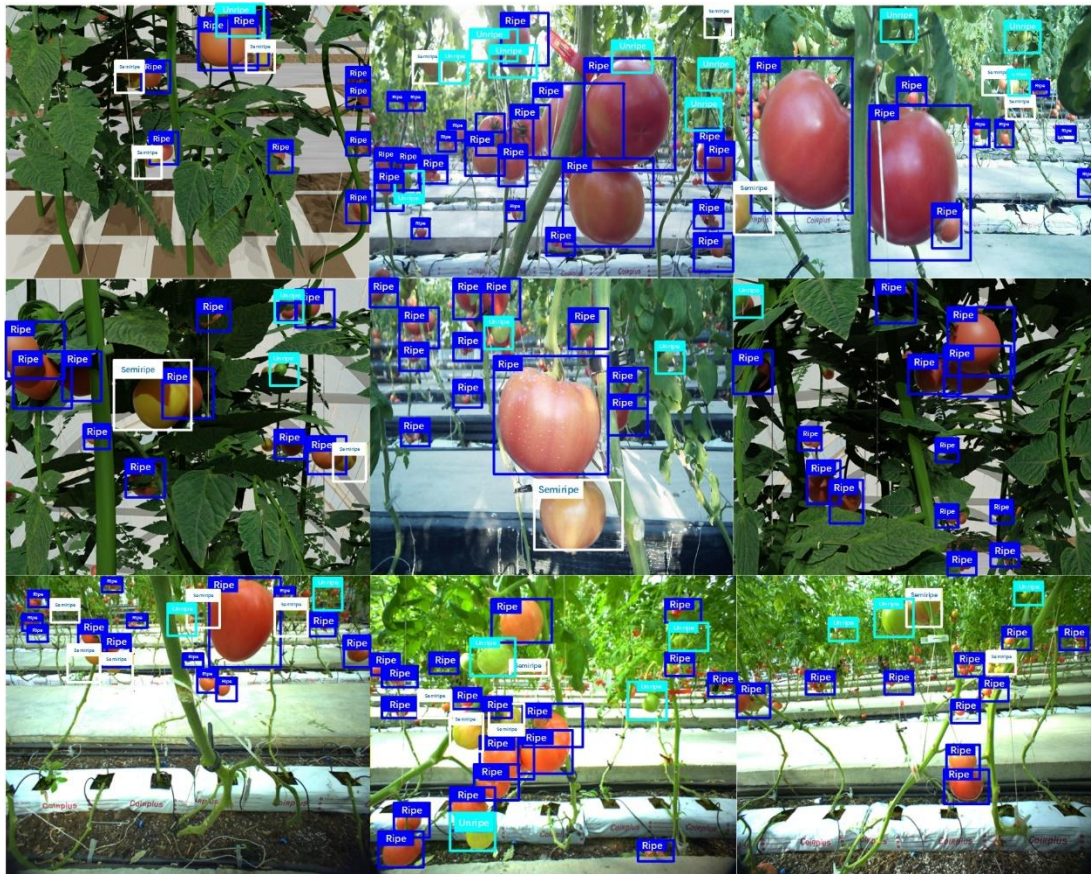


Fig. 15-(a)

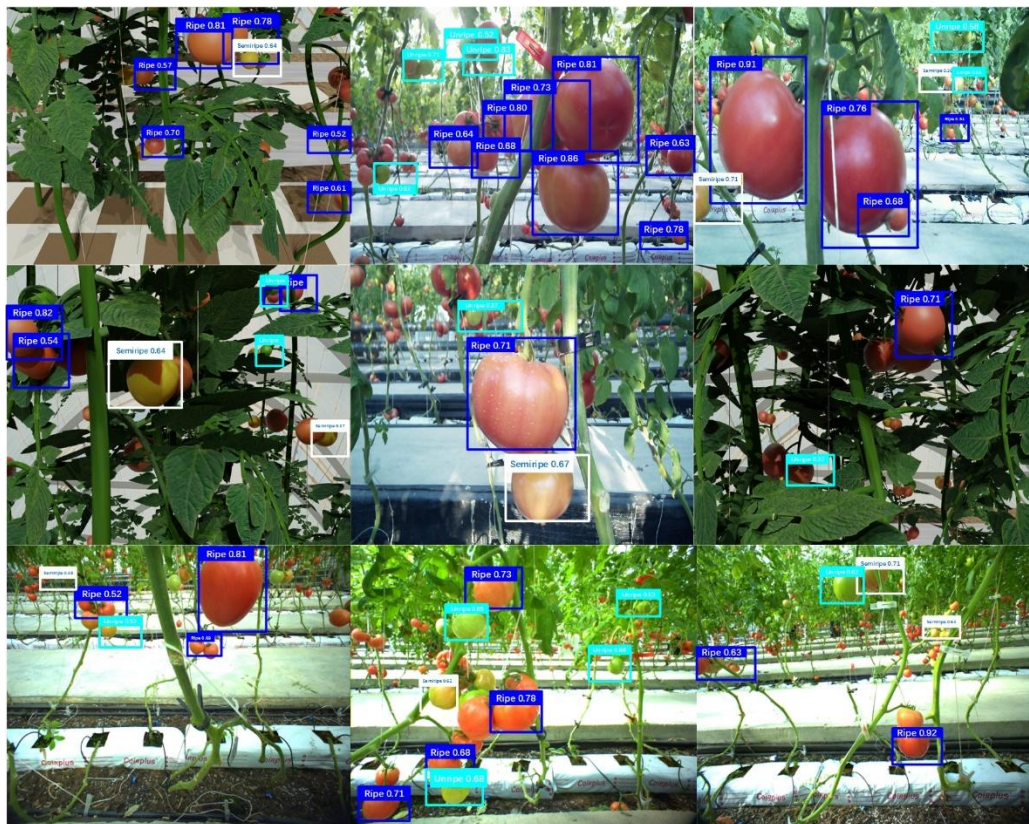


Fig. 15-(b)





integration of this detection model with agricultural robots, drones, or automated harvesting machinery to achieve a closed-loop system from “perception” to “execution.” By developing integrated hardware-software systems incorporating the model, an intelligent, precise, and efficient full-cycle management platform for tomato orchards can ultimately be established, providing stronger technical support for the advancement of smart agriculture.

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## **Data availability**

Data will be made available on request.

## **Declarations**

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## **Competing interests**

The authors have no relevant financial or non-financial interests to disclose.

## **Author contributions**

Fengyan Lun is the first author and corresponding author. The contributions of each author were listed using the CRediT taxonomy; conceptualization: Fengyan Lun; data curation: Fengyan Lun Chuang Wang.; investigation: Xiaolong Zhou, Yunsheng Tian; resources, and validation: Fengyan Lun Chuang Wang; formal analysis and visualization: Fengyan Lun, Xiaolong Zhou, Yunsheng Tian; writing – original draft: Fengyan Lun.; writing – review & editing: Fengyan Lun, Chuang Wang.

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